

Swarat Sarkar

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Summary

Software and AI engineer with 3+ years building production systems achieving >99.8% accuracy, serving 25k+ users, processing 30M+ daily transactions. Expert in Python, ML frameworks, and cloud platforms for end-to-end solutions.

Work Experience

Research Assistant | UB Safe and Efficient Autonomous Systems Lab | Buffalo, NY, USA Jul 2025 – Present

- Architecting **Computer Vision pipeline** in **ROS2** using **Vision Transformers** on autonomous vehicles for bridge crack segmentation and length quantification, enabling real-time monitoring through **vehicle-to-everything (V2X)** technology.
- Engineered **crack detection training pipeline** with custom augmentation strategies and **semi-automatic annotation tool** for **443 dashcam images**, improving F1-score from **0.29 to 0.55** and precision from **0.20 to 0.70** via adaptive cropping.
- Co-authored conference paper "Infrastructure-Guided Connectivity-Enhanced Road Crack Detection and Estimation" accepted for publication at **IEEE MOST 2026** (arXiv:2604.24616).

ML Engineer (Capstone Project) | Nissha Medical Technologies | Buffalo, NY, USA Sep 2025 – Dec 2025

- Pioneered **AI-driven Inspection System** using **YOLOv8s-TensorRT** achieving **>99.8% defect detection** with 84ms inference.
- Delivered **real-time quality dashboard** with root cause analysis for 30M+ daily casino tickets, addressing **4% defect rate** with **automated multi-gate inspection** (count, visibility, density, alignment), enabling waste reduction and scalable deployment.

Software Engineer | TATA STEEL Limited | Jamshedpur, India Apr 2023 – Aug 2024

- Engineered **Python-based analytics pipeline** processing **110k+ network incidents** using **Pandas, NumPy**, and statistical modeling, developing predictive insights that reduced outages by **40%** and decreased support call volumes by **25%**.
- Programmed **real-time monitoring platform** integrating **4 REST APIs (OPManager, HP Aruba, Cisco Meraki, ServiceNow)** with **Python backend** and **Tableau frontend**, delivering executive dashboards for **CIO weekly reports** and achieving 100% asset coverage across **16,000+ devices** with **99.5% uptime**.
- Developed **suite of automation tools** using **Python and Power Apps**, including bulk email dispatch, contract alerting, asset ID automation, and inventory management platform, streamlining workflows for **1,000+ users**.

Graduate Trainee - Software | TATA STEEL Limited | Remote Aug 2022 – Apr 2023

- Built a responsive web app using **ReactJS, ExpressJS, Google Maps API**, and **Microsoft SQL** to assist 25k+ users with **real-time bus tracking** and customized maps generated from routes, improving commute in the steel plant.
- Developed **JWT-based authentication** and password recovery system for internal .NET application, and devised **BigQuery pipeline** processing **75k+ user network connections** from GCP data lake for analytics workflows.

Technical Projects

Research Synthesis & Hypothesis Agent | *LangGraph, GPT-4o, Claude Sonnet, Pinecone, Supabase, FastAPI, AWS Lambda*

- Built a **7-node multi-agent research pipeline** with **cost-aware LLM routing** that autonomously fetches ArXiv papers, embeds via Pinecone, synthesizes findings (Claude), detects cross-paper contradictions (OpenAI), and generates **3 novel hypotheses**, achieving **~\$0.035 avg cost per query** and **~40s latency** logged to Supabase, deployed serverlessly on **AWS Lambda**.

Safety-Constrained Insulin Dosing RL Agent | *PyTorch, Stable-Baselines3, Gymnasium, Simglucose, Streamlit, Hugging Face*

- Built a **custom Gymnasium environment** over an **FDA-validated T1D simulator** and benchmarked **5 RL agents** (DQN, PPO, SAC + constrained variants) across **3 patient cohorts**, achieving **76.5% safe-range adherence**, matching clinician baseline, and recovering a collapsed PPO policy from **47% → 73%** via **Lagrangian safety constraints**.

Real-Time Soccer Offside Detection | *YOLOv8, OpenCV, Computer Vision*

- Developed offside-detection system converting YOLOv8 player/field boxes into geometric field lines using a custom **SVD-based vanishing-point solver**, and **HSV team classification**, achieving >20 FPS and **95% accuracy** on consumer GPUs.

Education

University at Buffalo, NY, USA | M.S Engineering Science – Artificial Intelligence | CGPA: 3.83/4.0 Aug 2024 – Feb 2026

Indian Institute of Technology Hyderabad, India | B.Tech Mechanical Engineering | CGPA: 8.1/10 Jul 2018 – May 2022

Skills

Programming & AI: C/C++, Python, MATLAB, Java, C#, R; Frameworks (TensorFlow, Keras, PyTorch, Gym, XGBoost); Deep Learning (CNNs, RNNs, Autoencoders); Reinforcement Learning (DQN, A3C, PPO); Generative AI (VAEs, Transformers, Diffusion)
Big Data & Data Science: Apache Spark (SQL, MLlib, GraphX), PySpark, Hadoop, Kafka; pandas, NumPy, matplotlib, scikit-learn
Cloud, DevOps & OS: GCP, AWS, Azure; Docker, Kubernetes; CUDA, TensorRT; Jupyter Notebook, VS Code, Git, Postman; Linux
IoT & Computer Vision: NVIDIA Jetson, Raspberry Pi, Arduino; OpenCV, ROS2, YOLO, CVAT; Visualization (Tableau, PowerBI)
Full-Stack and Databases: ReactJS, NodeJS, .NET, ExpressJS; MySQL, PostgreSQL, SQL Server, Oracle, MongoDB, Pinecone

Awards & Certifications

Awards: NASA Lunabotics 2025 - Best Presentation Award | Tata Steel - Suri Seva Foundation Award (2nd/200+ trainees)

Certifications: AWS Certified AI Practitioner, DeepLearning.AI TensorFlow Developer Specialization